

Benchmarking an open citation pipeline against CanLII's RefLex

Project status note, 28 September 2026

Summary

- CanLII's citator, RefLex, works out which decision each case-law citation refers to. According to CanLII's own descriptions, it switched in 2015 to the approach we reached independently: learning this from parallel citations printed together.
- We compared the two systems through the CanLII API on 360 judgments stratified by court and period. For recent judgments, we also find 94-97% of the cases CanLII lists as cited. For recent cited decisions, 210 of 211 citing decisions appear in both systems.
- Agreement drops for older material. We have not yet adjudicated all the differences, and they run in both directions.
- RefLex is a closed search tool. Our pipeline is open, keeps a record for every citation and measures its own errors, so it can serve as a research instrument, including for checking data taken from CanLII.

1. What RefLex is

The 2005 design

Poulin, Paré and Moganov (LexUM) described RefLex at Law via the Internet 2005. Editors in four law libraries compiled a table of about 200,000 citations from some 30 Canadian law reports published between 1985 and 2004, recording the court, date, docket number and style of cause of each decision. A program found citations in judgment text with regular expressions and looked each one up in the table. Citations missing from the table were recorded but got no link. A similarity score connected parallel citations of the same decision, and an editor decided the doubtful pairs. The authors also considered an automated, heuristic method that would not need the table. They set it aside because, depending on how it was built, it could produce many wrong links or very few links.

Reflex 3 (2015)

CanLII replaced the mainly manual system with an automated one. It adds citation entries by reading the parallel citations printed in headnotes and in the text of decisions. CanLII reported 18% more noting-up results. Editors reviewed more than 100 of the most important citation sources, and their corrections of conflicts feed back into automatic resolution.

Treatment and case history (2020)

CanLII added a treatment view that ranks citing cases by how much they discuss the cited case and flags "unfavourable mentions" found through marker words. It also automated judicial history. When words such as "affirmed" or "reversed" appear between two cited decisions, the system links them as the same litigation. It found about 28,000 candidate relations and accepted only those present in at least two citing documents, about 8,300.

CanLII and Lexum have not published algorithm details, parameters or accuracy figures for any version.

Sources:

- 2005 paper: <https://lexum.com/wp-content/uploads/2016/10/2005-reflex-lexum.pdf>
- 2015 post (Reflex 3): <https://blog.canlii.org/2015/03/17/canlii-professional-users-news/>

- 2020 post (treatment, history):

<https://blog.canlii.org/2020/11/17/a-new-wave-of-improvements-on-canlii-%F0%9F%8C%8A/>

2. Our pipeline in brief

The pipeline runs in five layers: extraction, classification, merging, adjudication and selection. Extraction recognizes a citation by its shape (year, volume, abbreviation, page) instead of checking a list of known abbreviations, so it picks up reports nobody listed in advance, from seventeenth-century English nominate reports to the United States Reports. The pipeline decides which decision a citation refers to from evidence in the corpus: neutral citations, the citations that corpus decisions print for themselves, co-citation, and case name with year. When that evidence is insufficient, the citation is recorded as undetermined. Every row can be traced to its position in the source text, and every known defect goes into a public problem ledger, which has 98 entries so far.

3. Side-by-side comparison

	RefLex (from public descriptions)	Our pipeline
Purpose	Hyperlinks and noting-up for practitioners	Auditable census of cited cases for research
Source of case identity	Editor-compiled table (2005); learned parallel-citation sets (2015 on)	Evidence in the corpus: neutral anchors, self-citations, co-citation, name and year
Unfamiliar abbreviations	Detected against a list of several hundred reports	Matched by structure; no abbreviation list
Citations it cannot resolve	Not linked, not reported	Kept, flagged, counted
Doubtful cases	An editor decides	Marked undetermined
Per-citation record	Not published	Position in text, printed form, rule applied
Error figures	None published	Problem ledger and external comparison (this report)
Reproducibility	Counts change as the system learns	A fixed corpus snapshot and rule version give identical output
Coverage	All Canadian courts and tribunals, plus legislation	Three courts so far
Treatment signals	Discussion intensity, unfavourable mentions	None

4. Cross-check against the CanLII API

Design

We sampled 40 judgments in each of nine strata (three courts by three periods), 360 judgments in all. Judgments from before neutral citation were mapped to CanLII identifiers by case name and year, and each sampled mapping was then checked against CanLII metadata: the decision date and, for the Supreme Court, the S.C.R. citation. For each judgment we compared CanLII's list of cited cases with ours. Separately, we compared the citing decisions of 60 cited cases, stratified by the age of the cited decision. The run used 689 API calls at three-second intervals, with responses cached locally. This report gives aggregate figures only.

Cited cases per judgment

Share of CanLII's cited cases that we also find, with 95% confidence intervals:

Court	Earliest period	Middle period	Latest period
SCC (1875-1949 / 1950-1999 / 2000-2026)	36% [28-46]	80% [76-84]	97% [96-98]
ONCA (1998-2006 / 2007-2015 / 2016-2026)	70% [61-78]	86% [79-90]	97% [95-98]
BCCA (1999-2007 / 2008-2016 / 2017-2026)	83% [77-88]	94% [91-96]	96% [94-97]

These figures measure agreement. Neither system is ground truth, and many non-matches are the same case under different identifiers: CanLII often gives only a CanLII citation where the judgment printed a law-report citation.

Citing decisions per cited case

Comparable citing decisions only:

Age of cited decision	CanLII's citing decisions we also find	Ours that CanLII also lists
1800-1949 (20 cases)	83%	98%
1950-1999 (20 cases)	89%	100%
2000-2026 (20 cases)	99.5% (210 of 211)	99.5%

Example: *Housen v. Nikolaisen*, 2002 SCC 33

Housen is the most-cited case in the corpus. Within the comparable range, the two systems share 1,457 citing decisions. Twelve citing decisions appear only in our data. We checked all twelve in the source text, and each is a genuine citation printed in a defective form that RefLex does not resolve, for example "2002 SCC 3", "2003 SCC 33", "[2002] SCC 35", "[2002] S.C.R. 235" with no volume, and "Hausen and Nikoleson". Three citing decisions appear only in CanLII's data, and all three are genuine as well. Two are our errors: one citation was given a borrowed case name and one carried a misprinted year. The third mentions *Housen* by name without a citation, which our extractor does not attempt to capture. We recorded the text position and a quotation for each of the fifteen.

5. What the comparison shows, and what it does not

Two systems built independently, one on an editor-compiled table and one on corpus evidence, give nearly the same answer for modern Canadian case law. We read this as good external support for our method.

The comparison also exposed errors on our side that we could only have found with a traceable pipeline:

- A trimming rule in case-name extraction cuts citation chains ("A v. B, citation, citing C v. D, citation") in the wrong place, so 4,455 citations (0.66% of citations with a case name) received the name of the preceding case. A random sample of 40 flagged rows contained 40 true errors.
- The neutral court code FCA belongs to both the Federal Court of Appeal of Canada and the Federal Court of Australia. Seventeen Australian citations were classified as Canadian.

It does not yet show that our pipeline is more complete or more accurate than RefLex overall:

- 318 citations that CanLII lists and we do not match still need adjudication by hand.
- For older periods, we have not explained the gap between the systems.
- About 2,300 older Ontario Court of Appeal judgments could not be mapped to CanLII identifiers and were left out of the comparison.